SYNOPSIS

Title

CineMatch – Personalized Movie Recommendation Engine

Introduction

Streaming platforms host vast, ever-growing content libraries, and user engagement depends heavily on how well relevant content is surfaced. This project builds a full-stack movie recommendation system that goes beyond simple genre/year filters — it learns from user rating patterns to suggest films tailored to individual taste, using a collaborative filtering approach served through a streaming-style web interface.

Problem Statement

Content streaming platforms rely on keeping users engaged with relevant content, but standard categorization by genre or release year is insufficient for surfacing hidden gems tailored to a user's specific, nuanced preferences — leading to poor content discovery and reduced watch-time.

Objectives

- Build a streaming-style web interface that collects and stores user movie ratings.
- Implement a collaborative filtering algorithm that identifies similarity between users (or items) based on rating patterns.
- Generate personalized movie recommendations and display them as dynamic UI carousels, similar to real streaming platforms.
- Continuously improve suggestions as more rating data is collected.

Scope

- Covers a fixed movie catalog (e.g., built on a public dataset such as MovieLens) with user-submitted ratings layered on top.
- Focuses on collaborative filtering (user-based and/or item-based); does not include video streaming/hosting itself.
- Designed as a single-platform web demo, not a production-scale multi-region streaming service.

Proposed Solution / Methodology

1. **Data Collection & Preprocessing** — Use a public ratings dataset (e.g., MovieLens) as the base catalog; build a schema for users, movies, and ratings; clean and normalize rating data.
2. **User Interface** — Streaming-style frontend where users browse a movie catalog and submit star ratings.
3. **Recommendation Engine (Model Dev)** — Implement collaborative filtering (matrix factorization / SVD, or user-item similarity via cosine/Pearson correlation) to predict ratings for unseen movies per user.

4. **Backend & API** — Serve recommendation results through a REST API that the frontend queries on load/refresh.
5. **Evaluation** — Validate predictions using standard metrics (RMSE/MAE on held-out ratings, precision@K for top-N recommendations).
6. **Presentation Layer** — Render personalized picks as horizontally scrollable "carousels" (e.g., "Recommended for You," "Because You Rated X").

Features / Modules

- User Authentication & Profile Module
- Movie Catalog Browser
- Rating Submission Module
- Collaborative Filtering Recommendation Engine
- Personalized Carousel Dashboard (frontend)
- Admin/Analytics Panel (optional — track engagement with recommendations)

Technology Stack

- **Language:** Python (backend/ML), JavaScript (frontend)
- **Backend:** FastAPI
- **Database:** PostgreSQL (users, movies, ratings)
- **ML Libraries:** scikit-learn / Surprise (for collaborative filtering, SVD)
- **Frontend:** Vanilla HTML/CSS/JS (or React, if a richer carousel UI is desired)
- **Dataset:** MovieLens (or similar public ratings dataset) for cold-start catalog data

System Requirements

- **Hardware:** Standard development machine (8 GB RAM minimum); GPU not required for classical collaborative filtering.
- **Software:** Python 3.10+, pip-managed dependencies (FastAPI, scikit-learn/Surprise, pandas).
- **Browser:** Any modern browser (Chrome/Firefox/Edge).
- **OS:** Windows 10/11 or Linux.

Expected Outcomes

- A working recommendation system that produces meaningfully personalized movie suggestions beyond basic genre filters.
- A functional streaming-style UI demonstrating real-time carousel-based recommendations.
- Measurable improvement in recommendation relevance (via RMSE/precision@K) as more user rating data accumulates.

Future Scope

- Hybrid recommendations combining collaborative filtering with content-based filtering (genre, cast, description embeddings).
- Cold-start handling for new users/movies with no rating history.
- Integration of implicit feedback (watch time, clicks) alongside explicit ratings.
- Deep learning–based recommenders (e.g., neural collaborative filtering) for improved accuracy at scale.

Conclusion

By combining collaborative filtering with a streaming-style interactive interface, this project delivers a personalized movie discovery experience that mirrors real-world platforms — showing how data-driven recommendations can meaningfully improve content engagement over static, genre-based browsing.